



Acute and sub-chronic oral toxicity studies of hesperidin isolated from orange peel extract in Sprague Dawley rats

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ABSTRACT

Citrus sinensis contains glycoside hesperetin-7-rhamnoglucoside (hesperidin) which harbor an array of therapeutic potentials including antioxidant, anticancer, and anti-inflammatory. However, a systematic examination of safety is needed before its utilization. Hence, the present investigation is aimed to evaluate acute and sub-chronic toxicity of hesperidin isolated from the citrus fruit. Hesperidin (73%) was isolated from a methanolic extract of dried peel of the citrus fruit, characterized using FTIR, and standardized by HPLC. Its acute oral toxicity (AOT) and sub-chronic toxicity studies were carried out in Sprague-Dawley rats. Hesperidin (5000 mg/kg) showed 10% mortality in AOT. In sub-chronic toxicity study, hesperidin (250 and 500 mg/kg) did not induce any abnormalities in body weight, food consumption, clinical signs, ophthalmological and neurological observations, urine analysis, hematology, clinical chemistry, organ weights, and gross pathology. However, hesperidin (1000 mg/kg) showed significant ($p < 0.05$) alterations in body and organ weights, hematology, clinical chemistry, and tissue histopathology. To conclude, hesperidin has median lethal dose (LD₅₀) of 4837.5 mg/kg, and Low Observed Adverse Effect Level (LOAEL) at 1000 mg/kg for both male and female Sprague-Dawley rats. Thus, hesperidin isolated from citrus fruit showed a good safety profile in animal study.

1. Introduction

Natural products are an essential source of drugs for management of various diseases. Numerous evidence have suggested that herbal therapeutic moieties are better in terms of safety and efficacy compared to synthetic chemicals (Aniagu et al., 2004; Delaney, 2007; Joshua Allan et al., 2007; Saad et al., 2006). Therefore, with the development of new medical technology, research on isolated compounds from natural and biological resources have garnered much attention. Various natural substances have wide applications in an array of fields including medicine, functional health food, and home remedies (Jakkula et al., 2004; World Health Organization, 2002; Yang et al., 2009).

In developing countries, majority of the population rely on folk medicine obtained from natural sources for the treatment of various diseases. However, these plants contain diverse bioactive compounds that bear the potential to cause an adverse effect (Bent and Ko, 2004). In the Indian subcontinent, 'Ayurveda' has been widely used as an 'alternative medicine' for the past 3000 years or so. Recently, Ayurveda has attained more focus in the field of medicine due to its safety and frequency of success. Moreover, it has been shown to provide excellent

clinical results with lower adverse effects than Western medicine (Corns, 2003; Mashour et al., 1998; Yin et al., 2013). But before these traditional medicines can be transformed into modern drugs, significant toxicological information is needed. However, in recent years, the toxicological studies of natural products have rarely been reported. Therefore, there is a need to conduct and document systematic safety studies which examine these substances for their possible toxic effects.

Citrus sinensis (L.) Osbeck, (family: Rutaceae), commonly known as Sweet orange, is a citrus fruit which is largely cultivated in India and various countries around the world. It serves as a major source of vitamin C, potassium, folic acid, and pectin. It is widely consumed as fresh fruit, and juice while the peel is often discarded as waste. According to a report by the United States Department of Agriculture, the global production of this citrus fruit is 82 million tons per year (United States Department of Agriculture, 2018) of which 34% is used for juice production which yields about 44% peel as a by-product (Rafiq et al., 2016). It is to be noted that these peels possess an array of secondary components including polyphenols, vitamins, amino acids, minerals, dietary fibers, essential oils, pectin, flavonoids, vitamin C, and carotenoids. Thus, these have a broad range of activities including

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antimicrobial, antioxidant, anticancer, diuretic, and stomachic (Grosso et al., 2013; Manthey and Grohmann, 2001; Wang et al., 2014). In traditional Chinese medicine, orange peels have been used as a tonic to treat disorders related to skin, immune system, digestive system, and vitamin deficiencies.

It has been reported that some of its promising biological properties can be attributed to the presence of flavonoids in citrus fruit peel extract (Rafiq et al., 2016). One of these is hesperidin (hesperetin-7-rhamnoglucoside) which is a flavonoid glycoside. The effects on experimental animals of various clinical and epidemiological studies have supported the role of hesperidin across multiple diseases. Hesperidin is reported to have therapeutic potential against various diseases including hyperlipidemia, diabetes, cancer, arthritis, infections, allergies, inflammation, hypertension, fibrosis, and oxidative stress (Akiyama et al., 2010; Jin et al., 2008; Kandhare et al., 2017; Kaur et al., 2006; Kawaguchi et al., 1997a; Lee et al., 2004; Shah and Patel, 2010; Shi et al., 2012). Hesperidin has been known to play a vital role in the prevention of diabetes-induced complications including nephropathy, neuropathy, cardiomyopathy, and brain damage (Ibrahim, 2008; Shah and Patel, 2010; Visnagri et al., 2014). Previous research showed that treatment with hesperidin attenuates sodium arsenite-induced toxicity (Pires Das Neves et al., 2004) and breast cancer in rats (Nandakumar et al., 2011). Hence, hesperidin isolated from citrus peel extracts received considerable attention for the management of various diseases. In view of the above, use of hesperidin for clinical management of various diseases is expected to increase. However, no cases of intoxication have been reported to date, and no adverse effect have been systematically studied after hesperidin ingestion.

Various animal models have played a significant role in the evaluation of safety of drugs and plant products (Delaney, 2007). Therefore, toxicological insult in rat and mice provide a good correlation of safety between the preclinical and clinical settings (Olson et al., 2000). In the past, various studies have been undertaken with high doses in mice for acute effects, and lower doses in rats for chronic effects to determine the clinically useable dose of potential drugs while abiding by the guidelines of Organization for Economic Cooperation and Development (OECD) (Jadeja et al., 2011; OECD, 1998; OECD, 2008). To our knowledge, toxicological evaluation of hesperidin isolated from citrus fruit extract has not been systematically studied yet. Hence, the present study was aimed to investigate the toxicological effects, if any, of hesperidin isolated from citrus peel extracts in rats by acute and sub-chronic administration using biochemical, hematological, and histopathological examinations in accordance with the OECD guidelines.

2. Materials and methods

2.1. Collection and authentication

Fresh fruits of *Citrus sinensis* (L.) Osbeck, Rutaceae, were collected between November and December 2013 at Nagpur, India. These were authenticated by the Department of Botany, Agharkar Research Institute, Pune. Voucher specimen of plant materials are maintained in the laboratory. Macroscopic characteristics of the fresh and dried peels were noted. Chopped green peels were dried for 36 h in the sunlight, followed by shade drying. Dried peels were powdered in a mixer grinder, packed in paper bags, and stored in an airtight container until use.

2.2. Extraction and isolation of hesperidin

The extraction and isolation of hesperidin were carried out according to the previously reported method (Al-Ashaal and El-Sheltawy, 2011; Bigoniya and Singh, 2014). Briefly, the dried powdered peel was placed in a reflux condenser with petroleum ether for 1 h, filtered, dried at room temperature, and the yield was calculated. The methanol extract was subjected to phytochemical evaluation for alkaloids, tannins,

phenolics, saponins, carbohydrates, steroids, flavonoids, and glycosides according to a previously reported method (Kokate et al., 2001). The peel extract was re-extracted with methanol for 3 h, filtered, and washed with hot methanol. The filtrate was concentrated and dried. The dried residue was mixed with 50 mL of 6% acetic acid to precipitate crude hesperidin. Crude hesperidin was filtered using a Buchner funnel, washed with 6% acetic acid, and dried to constant weight.

2.3. Purification of hesperidin

Dimethylformamide, added to crude hesperidin (7 mL/g), was prepared by warming it to 60 °C and treated for 30 min. With activated charcoal previously boiled with dilute hydrochloric acid. Glacial acetic acid was added to the formamide solution to make it slightly acidic. The solution was then filtered through a Celite filter, diluted with an equal volume of water, and allowed to stand for a few hours in order to crystallize. The crystals of hesperidin were filtered, washed with hot water, and the percentage yield was calculated. The isolated hesperidin was subjected to ferric chloride test and magnesium-hydrochloric acid reduction test (Shinoda test). The melting point of isolated hesperidin was determined.

2.4. Identification of hesperidin

1 mL of Hesperidin stock solution was diluted to 10 mL with methanol in order to get a solution of 50 µg/mL, and UV spectra were assessed with a spectrophotometer (UV-visible spectrophotometer, Jasco V-530, Tokyo, Japan). The FT-IR spectra (Fourier-transform infrared spectroscopy, PerkinElmer FT-IR spectrophotometer) were recorded using KBr pellet at a scanning speed of 2 mm/s and with the resolution set at 4 cm⁻¹.

2.5. HPLC of hesperidin

The HPLC analysis was carried out on Chrompass liquid chromatographic-4000 system (Jasco, Tokyo, Japan) equipped with auto-sampler and UV-2075 plus detector. Data acquisition was controlled by JASCO ChromPass chromatography data system software (JASCO GmbH, Grob-Umstadt, Germany). 20 µL sample solution (50 µg/mL) was injected and chromatographic separation was accomplished on a Eurospher C18 (250 mm × 4.0 mm × 5 µm) analytical column (Sigma-Aldrich, Mumbai, India) with pump (PU-2080 plus), UV detector (UV-2075), and auto-sampler (AS-2055 plus). The mobile phase was 5 mM ammonium acetate/Acetonitrile 75:25 (v/v) adjusted to pH 4.45 with acetic acid at a flow rate of 1 mL/min. Chromatograms were monitored at 285 nm with a runtime of 30 min, and the column temperature was maintained at 25 °C.

2.6. Animals

Sprague Dawley rats (both sexes and 6–8 weeks old, 190–210 g) were obtained from National Institute of Biosciences, Pune (India). The females were nulliparous and non-pregnant. The rats were kept for seven days for acclimatization prior to dosing, cages were marked, and individual marking was also made on fur for identification. They were maintained at 24 °C ± 1 °C, with a relative humidity of 45–55% and 12:12 h dark/light cycle. The animals had free access to standard pellet chow (Pranav Agro-industries Ltd., Sangli, India) and water throughout the experimental protocol. All experiments were carried out between 0900 and 1700 h. The experimental protocol was approved by the Institutional Animal Ethics Committee (IAEC) of Poona College of Pharmacy, Pune and performed in accordance with the guidelines of the Committee for control and Supervision of Experimentation on Animals (CPCSEA).

2.7. Acute studies

The acute toxicity study of hesperidin was performed according to OECD guideline No. 425 (OECD, 2008). Ten healthy rats per treatment group (5 males and 5 females) were used in this study. The dose was freshly formulated with distilled water as the vehicle. The dose volume was kept at 10 mL/kg, and a dose of hesperidin (55, 175, 550, 1750, and 5000 mg/kg) was administered orally. Each was a single ordered dose progression in which animals were dosed, one at a time, at 48-h intervals. The first animal received a dose a step below the level of the best estimate of LD₅₀ (Hamilton et al., 1977). If the animal survived, the dose for the next animal was increased to a factor of one half log times the original dose; if it died, the dose for the next animal was decreased in a similar dose regression. The vehicle control (distilled water, 10 mL/kg) was also maintained as a separate group. The animals were observed for 14 days for any signs of morbidity or mortality.

2.8. Sub-chronic toxicity studies

The sub-chronic toxicity study of hesperidin was performed according to OECD guideline No. 408 (OECD, 1998). Rats were randomly divided into various groups of 15 animals per sex. They received a dose of hesperidin (0, 250, 500, and 1000 mg/kg) for 13-weeks. A separate group of rats, treated with hesperidin (0 mg/kg (Rev.) and 1000 mg/kg (Rev.)), were also maintained for reversal dosing with 28 days of reversal period. The doses of hesperidin were freshly prepared daily in distilled water and administered to the rats by oral gavage for 13-weeks.

2.8.1. Clinical signs, body weight, and feed

All animals were observed daily for mortality. The weight of each rat was weekly recorded throughout the study period. The consumption of feed and water were also observed.

Ophthalmoscopic evaluation, i.e., eye examination, was carried out using an ophthalmoscope (HEINE mini 2000, D-0171119, 1997) after induction of mydriasis with 0.5% solution of tropicamide.

Functional observation began with brief home cage observations, and the observer described the posture, palpebral closure, and presence of convulsions or tremors in the rats. The rat's reactivity to removal from the cage and handling were then rated, and the observer noted and/or ranked the presence of salivation, lacrimation, piloerection, and ptosis. Then, the rat was placed on an open field (the top of a 60 X 90-cm laboratory cart) for 3 min, and the presence of convulsions or tremors was again noted. Reactions to the approach of a pencil, touch on the rump, finger snap, and tail pinch using forceps were noted. The pupillary response was tested using a penlight stimulus. Righting reflex, or ability to right in midair, was noted.

2.8.2. Clinical laboratory investigations

After placing individual animals in metabolic cages (Techniplast, Italy), urine samples were collected for urinalysis from all animals in the 13-weeks study and the recovery groups during the last week of the treatment period, prior to the scheduled necropsy.

Immediately after urine collection, blood was collected from the animals by ROP (retro-orbital puncture) under ethereal anaesthesia and centrifuged at 7500 rpm for 15 min at 4 °C. Then, the serum was transferred to Eppendorf tubes using a micropipette and stored at 4 °C till analysis. For hematological investigations like Hemoglobin (Hb), Red Blood Corpuscles (RBC), Reticulocytes (Rt), Hematocrit (HCT), Mean Corpuscular Volume (MCV), Mean Corpuscular Hemoglobin (MCH), Mean Corpuscular Hemoglobin Concentration (MCHC), Total Leukocyte count (TLC), and percentage of cells in differential Leukocyte count (DLC), parameters like Neutrophils (N), Lymphocytes (L), Eosinophils (E), Monocytes (M), and Basophils (B) were determined. Similarly, serum biochemistry parameters, such as Blood Urea Nitrogen (BUN), Alanine Aminotransferase (ALT), Aspartate Aminotransferase

(AST), Alkaline Phosphatase (ALP), Gamma Glutamyl Transferase (GGT), Creatine Phosphokinase (CPK), Lactate Dehydrogenase (LDH), Blood Sugar (Sug), Calcium (Ca), Phosphorus (P), Bilirubin (Bil), Albumin (Alb), Creatinine (Creat), Sodium (Na), Potassium (K), Chlorine (Cl), Cholesterol (Chol), and Triglycerides (Trigly), were assayed using commercially available reagent kits according to the manufacturer's protocol (Accurex Biomedical Pvt. Ltd., Mumbai, India).

2.8.3. Necropsy, organ weight, and histopathology

On completion of the treatment, necropsy was performed on animals. Organs were isolated to determine their weights. Histopathology was performed on the isolated organs like adrenal gland, brain, colon, heart, kidney, lung, liver, ovary, pancreas, stomach, spleen, and testis of rats for their physiological characteristics. For histopathological evaluation, the tissues were fixed with 4% buffered paraformaldehyde solution and embedded in paraffin. Three to 4- μ m thick paraffin sections were dewaxed and brought to water through graded ethanol. Sections were stained with hematoxylin-eosin stain (H & E), dehydrated through graded ethanol, cleared in xylene, and mounted with Distyrene Plasticizer Xylene. H & E stained sections were graded to verify morphological assessment of tissue damage. Photomicrographs were captured at 40X magnification.

2.9. Statistical analysis

Data were represented as the mean $\pm$ standard error of the mean (SEM). Data analysis was performed using GraphPad Prism 5.0 software (GraphPad, San Diego, CA, USA). Data were analyzed using one-way ANOVA followed by Dunnett's test for each parameter separately. Data of 'mortality' was analyzed by Fisher's exact test. A value of $p < 0.05$ was considered statistically significant.

3. Results

3.1. Isolation, and characterization of hesperidin

The methanol extract of citrus peels showed the presence of carbohydrates, glycosides, steroids, tannins, flavonoids, and polyphenols but absence of alkaloids. The percentage yield of methanolic extract was 60.24%. Isolated hesperidin is a light yellowish amorphous powder (yield 50.24%). With the ferric chloride test, it showed wine red color whereas, for Shinoda test it gave bright pinkish violet color. The melting point of isolated hesperidin was found to be 242–244 °C.

In the identification by UV, absorption maxima (λ_{max}) of isolated hesperidin was 284 nm (Supplementary Fig. 1). The FR-IR spectrum of isolated hesperidin, showed in Supplementary Fig. 2, depicted a strong band of OH str. at 3640 cm⁻¹, CH arene at 3072 cm⁻¹, C–H str. (alkane) at 2995 cm⁻¹, C–H str. (alkane-OCH₃) at 2896 cm⁻¹, C=O str. at 1675 cm⁻¹, and C=C aromatic at 1625 cm⁻¹. HPLC chromatogram showed retention time at 5.82 min and 5.80 min for standard and isolated hesperidin, respectively (Fig. 1). The purity of isolated hesperidin was found to be 73%.

3.2. Acute toxicity study

Administration of the vehicle or hesperidin (55, 175, 550, and 1750 mg/kg) in both male and female rats did not produce any signs of intoxication, and all animals survived through the study period of 14 days. However, administration of hesperidin (5000 mg/kg) showed 10% mortality on the 11th day of the post-dosing recovery period as compared to vehicle control group. The acute toxicity data obtained by Karber's method indicated that the calculated median lethal dose (LD₅₀) value for hesperidin was 4837.5 mg/kg (Table 1).

When compared with vehicle control group, the body weight of male and female rats did not differ significantly in hesperidin treated rats after 14 days. Hesperidin did not show any significant alterations in

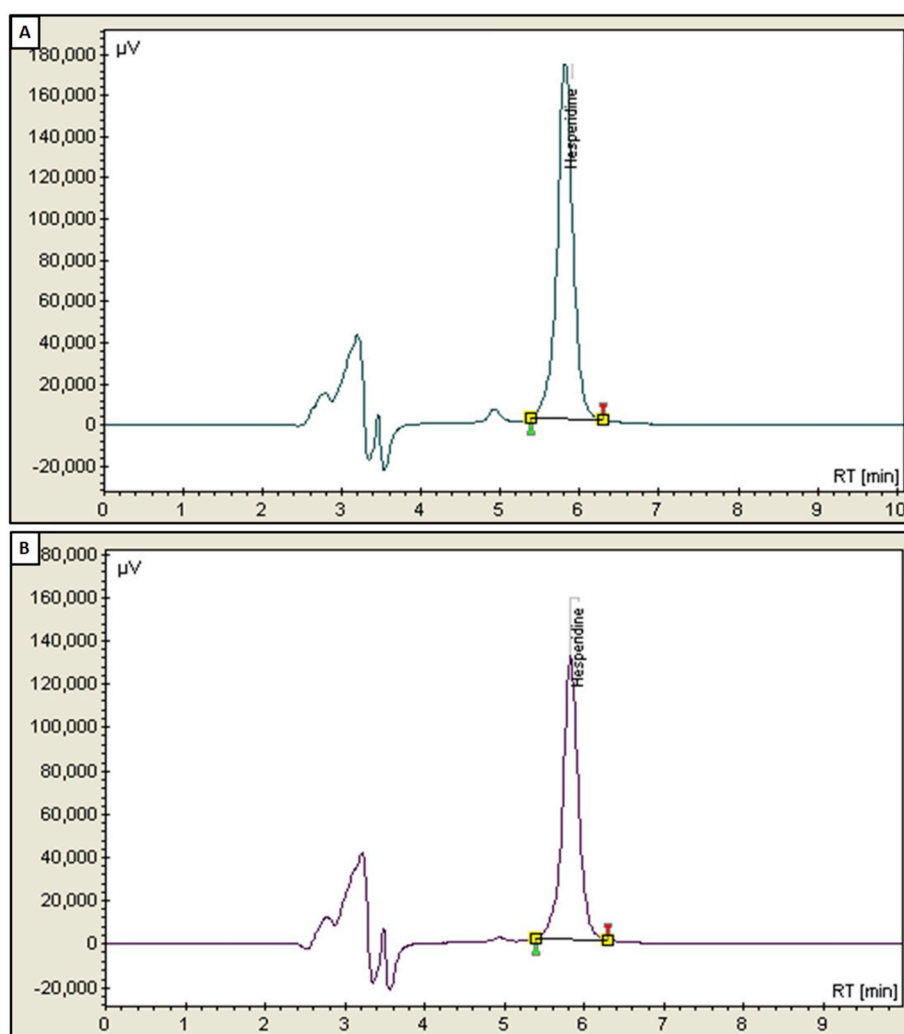


Fig. 1. HPLC chromatogram of standard hesperidin (RT = 5.82 min) (A) and isolated compound hesperidin (RT = 5.8 min) (B).

food intake and water intake when compared with vehicle control rats. Gross pathology findings and macroscopic examination of animals did not show any abnormalities in hesperidin treated as well as vehicle control rats. However, there was a significant increase ($p < 0.01$ and $p < 0.05$) in absolute liver and spleen weight of hesperidin treated (5000 mg/kg) rats as compared to vehicle control rats (Supplementary Table 1 and Supplementary Table 2).

Relative liver weight increased significantly ($p < 0.01$) in hesperidin treated (5000 mg/kg) male and female rats whereas relative spleen

weight increased significantly ($p < 0.05$) in hesperidin treated (5000 mg/kg) female rats as compared to vehicle control rats (Supplementary Table 3 and Supplementary Table 4).

3.3. Sub-chronic toxicity study

3.3.1. Survival, clinical signs, body weights

All animals from the vehicle control and the treated dose groups survived throughout the dosing period of 13-weeks and the post-dosing

Table 1

Effect of hesperidin on the survival of Sprague-Dawley rats during Acute Oral Toxicity (AOT) study.

Treatment (dose in mg/kg)	Sex	Dead/Total	Dead (%)	Latency	Symptoms
(1) Hesperidin (0)	(2) Male	(3) 0/5	(4) 0	(5) –	(6) None
	(8) Female	(9) 0/5	(10) 0	(11) –	(12) None
(13) Hesperidin (55)	(14) Male	(15) 0/5	(16) 0	(17) –	(18) None
	(20) Female	(21) 0/5	(22) 0	(23) –	(24) None
(25) Hesperidin (175)	(26) Male	(27) 0/5	(28) 0	(29) –	(30) None
	(32) Female	(33) 0/5	(34) 0	(35) –	(36) None
(37) Hesperidin (550)	(38) Male	(39) 0/5	(40) 0	(41) –	(42) None
	(44) Female	(45) 0/5	(46) 0	(47) –	(48) None
(49) Hesperidin (1750)	(50) Male	(51) 0/5	(52) 0	(53) –	(54) None
	(56) Female	(57) 0/5	(58) 0	(59) –	(60) None
(61) Hesperidin (5000)	(62) Male	(63) 1/5	(64) 20	(65) > 10 days, < 12 days	(66) Hypoactivity, Asthenia
	(68) Female	(69) 0/5	(70) 0	(71) –	(72) None

Data of “mortality” was analyzed by Fisher's exact test. Hesperidin was administered orally only once on day 0.

recovery period. No clinical signs of toxicity were observed in any of the groups during the entire study period and the 28 days of reversal period. Ophthalmoscopic examination did not reveal any abnormality for both the male and female rats at dose level 0–1000 mg/kg. Functional observation tests conducted on animals from the vehicle control and 1000 mg/kg dose group in week 13 revealed no abnormalities attributable to the treatment. Necropsy also did not reveal any abnormality attributable to the treatment.

There was no significant difference in the body weight of both male and female hesperidin (250 and 500 mg/kg) treated rats as compared to vehicle control rats at the end of the dosing period of 13 weeks. However, significant ($p < 0.01$ and $p < 0.05$) body weight gain in hesperidin (1000 mg/kg) treated male as well as female rats were found as compared to vehicle control rats. During the post-dosing recovery period on day 119, a significant increase ($p < 0.05$) in body weight was exhibited by hesperidin (1000 mg/kg-R) treated male and female rats as compared to vehicle control rats (Supplementary Table 11 and Supplementary Table 12).

3.3.2. Clinical pathology- hematological investigations

Hematological investigations conducted at the end of dosing period on week 13 and the end of recovery period, revealed significant changes in the values of different parameters studied when compared with that of respective vehicle controls. However, the increase/decrease in the values obtained was within normal biological and laboratory limits or the effect was not dose-dependent. In males, Hb, and HCT values significantly increased ($p < 0.01$) in hesperidin (1000 mg/kg) treated as well as reversal dose group, and RBC count also significantly increased ($p < 0.01$ and $p < 0.05$) in hesperidin (1000 mg/kg) treated and reversal dose group as compared to the respective vehicle control groups. The MCHC value significantly decreased ($p < 0.05$ and $p < 0.01$) in hesperidin (1000 mg/kg) treated and reversal dose group. Animals from hesperidin (1000 mg/kg) reversal dose group showed a significant increase ($p < 0.05$) in the platelet count as compared to vehicle control animals. Total WBC values significantly increased ($p < 0.01$) in hesperidin (1000 mg/kg) group as compared to vehicle control animals (Table 2) (Supplementary Table 5).

In females, the hesperidin (1000 mg/kg) treated and reversal dose group showed a significant increase ($p < 0.05$) in Hb count and RBC count also increased significantly ($p < 0.05$) in hesperidin (1000 mg/kg) treated as compared to vehicle control group. The HCT count significantly increased in hesperidin (1000 mg/kg) treated group whereas MCHC significantly decreased ($p < 0.01$ and $p < 0.05$) in hesperidin (1000 mg/kg) treated group as well as reversal dose group when compared with vehicle control group. Total WBC significantly decreased in ($p < 0.05$) hesperidin (1000 mg/kg) treated group as

compared to vehicle control group (Table 2) (Supplementary Table 6).

3.3.3. Blood chemistry

In males, there was a significant decrease ($p < 0.01$) in the level of blood urea nitrogen, blood sugar, and chloride whereas significant increase ($p < 0.05$) in the level of alanine aminotransferase in hesperidin (1000 mg/kg) treated animals as compared to vehicle control animals was observed. Calcium and sodium levels significantly increased ($p < 0.01$) in male animals of hesperidin (1000 mg/kg) treated and reversal group. In case of female rats, the calcium and sodium levels increased significantly ($p < 0.05$) in hesperidin (1000 mg/kg) reversal groups as compared to vehicle control. Rats dosed with hesperidin (1000 mg/kg) showed a significant increase ($p < 0.01$ and $p < 0.05$) in aspartate aminotransferase and triglycerides when compared with vehicle control group. The level of potassium decreased significantly ($p < 0.05$) in hesperidin (1000 mg/kg) dosed group as compared to vehicle control group (Table 3, Supplementary Table 7 and Supplementary Table 8). However, the increase/decrease in the values of blood chemistry parameters was within normal biological and laboratory limits or the effect was not dose-dependent.

3.3.4. Urine analysis

Urine analysis did not reveal any significant difference between both male as well as female rats administered with hesperidin (250, 500 and 1000 mg/kg) and hesperidin (1000 mg/kg) reversal group when compared with vehicle control group (Supplementary Table 9 and Supplementary Table 10).

3.3.5. Food consumption and organ weight

There was no significant difference in the food consumption of hesperidin treated rats during the 13-weeks dosing period and the post-dosing recovery period when compared with vehicle control animals (Supplementary Table 11 and Supplementary Table 12).

The weights of liver and spleen increased significantly ($p < 0.05$) in hesperidin (1000 mg/kg) and reversal group in male rats as compared to respective vehicle controls. In females, weight of liver ($p < 0.01$ and $p < 0.05$) and spleen ($p < 0.05$) increased significantly in hesperidin (1000 mg/kg) and reversal group as compared to respective vehicle controls as well. The weight of heart significantly increased ($p < 0.05$) in hesperidin (1000 mg/kg) treated male and female rats as compared to respective vehicle controls.

When compared with vehicle control rats, relative kidney weight decreased significantly ($p < 0.05$) in male rats administered with hesperidin (1000 mg/kg). Administration of hesperidin (1000 mg/kg) showed a significant decrease ($p < 0.05$ and $p < 0.01$) in the relative kidney weight of females as well at week 13 and reversal period

Table 2

Effect of hesperidin on hematological parameters of Sprague-Dawley rats during 13-weeks repeated dose toxicity study.

Parameter	H (0)	H (0)-R	H (1000)	H (1000)-R
Male				
Hb (g%)	13.93 ± 0.38	15.27 ± 0.56	16.45 ± 0.56**	16.43 ± 0.37**
RBC (x10 ⁶ /μL)	13.83 ± 0.31	15.33 ± 0.61	16.5 ± 0.50**	16.33 ± 0.61*
HCT (%)	40.00 ± 0.52	40.67 ± 0.21	42.17 ± 0.60**	42.00 ± 0.45**
MCHC (%)	36.00 ± 0.37	36.17 ± 0.60	33.50 ± 0.56**	33.83 ± 0.60*
Platelets (x 10 ³ /μL)	405.70 ± 5.39	400.50 ± 10.41	409.20 ± 7.87	438.50 ± 8.77*
TLC (x 10 ³ /μL)	10.83 ± 0.40	10.50 ± 0.22	11.00 ± 0.45**	12.67 ± 0.42
Female				
Hb (g%)	14.92 ± 0.51	15.65 ± 0.41	16.25 ± 0.67*	16.35 ± 0.55*
RBC (x10 ⁶ /μL)	15.00 ± 0.58	15.67 ± 0.49	16.33 ± 0.56*	15.00 ± 0.37
HCT (%)	40.67 ± 0.33	40.00 ± 0.37	42.17 ± 0.31**	39.67 ± 0.33
MCHC (%)	36.50 ± 0.56	36.17 ± 0.31	33.33 ± 0.33**	33.83 ± 0.54*
TLC (x 10 ³ /μL)	11.17 ± 0.31	11.00 ± 0.37	9.33 ± 0.21*	11.17 ± 0.40

Data are expressed as Mean ± Standard Error of Mean and analyzed by one-way ANOVA followed by Dunnett's test for each parameter separately. * $p < 0.05$, ** $p < 0.01$ as compared to respective parameter value of vehicle control (VC) i.e. Hesperidin (0 mg/kg). Hb: Hemoglobin; RBC: Red Blood Corpuscles, HCT: Hematocrit, MCHC: Mean Corpuscular Hemoglobin Concentration, TLC: Total leukocyte (White Blood Corpuscles) count.

Table 3

Effect of hesperidin on Blood Chemistry of Sprague-Dawley rats during 13-weeks repeated dose toxicity study.

Parameter	H (0)	H (0)-R	H (1000)	H (1000)-R
Male				
BUN (mg%)	34.63 ± 0.86	33.79 ± 0.95	29.98 ± 0.84**	36.12 ± 0.85
ALT (IU/L)	34.95 ± 0.96	40.16 ± 1.64	41.64 ± 0.58*	35.5 ± 1.25
Bl. Sug (mg%)	96.82 ± 1.63	96.48 ± 2.05	93.76 ± 1.88*	96.63 ± 1.84
Ca (mg%)	2.27 ± 0.10	2.14 ± 0.08	2.63 ± 0.06*	2.69 ± 0.06*
Na (mmol/l)	142.20 ± 0.39	141.20 ± 0.36	145.20 ± 0.66*	145.20 ± 0.77*
Cl (mmol/l)	97.05 ± 1.58	98.96 ± 1.29	89.95 ± 1.32**	97.45 ± 1.28
Female				
AST (IU/L)	65.32 ± 0.73	64.89 ± 0.86	70.58 ± 0.89**	65.42 ± 0.73
Ca (mg%)	2.29 ± 0.09	2.28 ± 0.11	2.22 ± 0.11	2.63 ± 0.12*
Na (mmol/l)	141.30 ± 0.79	141.40 ± 0.80	141.20 ± 0.65	145.00 ± 0.73*
K (mmol/l)	3.92 ± 0.06	3.93 ± 0.05	3.55 ± 0.13*	4.04 ± 0.08
Trigly (mg%)	105.40 ± 0.48	105.70 ± 0.37	106.60 ± 0.41*	106.30 ± 0.72

Data are expressed as Mean ± Standard Error of Mean and analyzed by one-way ANOVA followed by Dunnett's test for each parameter separately. * $p < 0.05$, ** $p < 0.01$ as compared to respective parameter value of vehicle control (VC) i.e. Hesperidin (0 mg/kg). BUN: Blood Urea Nitrogen, ALT: Alanine Aminotransferase, AST: Aspartate Aminotransferase, Bl. Sug: Blood Sugar, Ca: Calcium, Na: Sodium, K: Potassium, Cl: Chlorine, Chol: Cholesterol, Trigly: Triglycerides.

compared to vehicle control rats. In male and female rats, the relative heart weight decreased significantly ($p < 0.01$) in hesperidin (1000 mg/kg) as compared to vehicle control group. When compared with vehicle control group, relative thymus weight of males decreased significantly ($p < 0.01$) in hesperidin (1000 mg/kg) and reversal group, and in females also the relative thymus weight decreased significantly ($p < 0.01$ and $p < 0.05$) in hesperidin (1000 mg/kg) and reversal group (Supplementary Table 11, Supplementary Table 12, Supplementary Table 13 and Supplementary Table 14).

3.3.6. Histopathology

Histopathological findings of hesperidin treated animals are summarized in Table 4. The histopathological findings of adrenals, brain, heart, kidney, pancreas, spleen, testes and ovary from hesperidin (1000 mg/kg) treated animals did not show any histological aberrations (Supplementary Fig. 3 and Supplementary Fig. 4). However, there was mild inflammatory infiltration (red arrow) and necrosis (white arrow) in the colon (Fig. 2B and Fig. 2D) and minimal focal to multifocal lymphocytic infiltration (red arrow) in the liver (Fig. 2F and H) was observed in hesperidin (1000 mg/kg) treated animals. The histopathological examination of lungs from hesperidin (1000 mg/kg) administered rats revealed the presence of moderate inflammatory cells (red arrow) and mild necrosis (white arrow) (Fig. 2J and L) whereas minimal inflammatory cells (red arrow) and necrosis (white arrow) was observed in the stomach (Fig. 2N and P).

4. Discussion

In developing countries, herbal medicines are the cynosure of

Table 4

Effect of hesperidin on incidence and severity of histopathological findings of the animals.

Sex	Males		Females	
Hesperidin (mg/kg/day)	H (0)	H (1000)	H (0)	H (1000)
Number of animals	5	5	5	5
Colon				
Inflammatory cells	0	2	0	2
Necrosis	0	2	0	2
Liver				
Inflammatory cells	0	2	0	2
Lungs				
Inflammatory cells	0	3	0	2
Necrosis	0	2	0	0
Stomach				
Inflammatory cells	0	2	0	2
Necrosis	0	3	0	0

scientific attraction due to their wide use as alternative medicines. However, these herbal remedies significantly affect the clinical outcomes when administered as a supplement with others, thus needing more careful monitoring (Stanley et al., 2005). In view of the increasing use of hesperidin as a pharmacological intervention for potential health benefits, and with the dearth of its toxicological documentation, there is a pressing need to clarify the toxicological profile of hesperidin. In the present investigation, we presented a comprehensive safety evaluation of hesperidin isolated from citrus peel, by performing its acute (14-days) and sub-chronic oral toxicity (13-week) studies in Sprague-Dawley rats.

Hesperidin (hesperetin-7-rhamnoglucoside), isolated from dried peels of *Citrus sinensis* (L.) fruits, exhibits a melting point of 242–244 °C. It has been well documented that natural hesperidin is normally extracted with its chalcone and depending on isolation conditions, the melting point of hesperidin can vary from 243 to 257 °C (El-Nawawi, 1995; Hendrickson and Kesterson, 1954). In contrast, chemically synthesized colorless hesperidin crystal exhibits a melting point of 261–262 °C. In this regard, the result of the present investigation was in accordance to the findings of previous researchers that documented similar melting point of hesperidin (El-Nawawi, 1995; Hendrickson and Kesterson, 1954).

In the present study, exposure of rats to a single dose of hesperidin (up to 1750 mg/kg) did not produce any mortality and alterations in body weight, adverse clinical signs, and food consumption during 14 days of the acute oral toxicity testing. The result of the present investigation is in line with findings of previous investigators where administration of hesperidin (10% diet, for 30 days) did not affect the daily food intake (Kawaguchi et al., 1997b). It has been documented that, during 14 days of the acute oral toxicity testing, if the substance did not produce any clinical signs of toxicity or cause death at a dose of 5 gm/kg, then it can be considered to be non-toxic (Qin et al., 2009). However, hesperidin (5000 mg/kg, i.e., 5 g/kg) showed 10% mortality on the 11th day of post-dosing. Hence, we conclude that MLD (Minimum Lethal Dose) of hesperidin is more than 5000 mg/kg. The calculated median lethal dose (LD₅₀) value for hesperidin was 4837.5 mg/kg. Furthermore, in view of mortality during 14 days, male rats were found to be more sensitive towards hesperidin than females. It has been documented that, males have significantly longer gastric residence time which delayed absorption with a meal, hence they face a greater risk for clinically relevant adverse drug reactions (Anderson, 2008). As a flavonoid, hesperidin has been reported to have delayed absorption, thus resulting in delayed gastric emptying time in a male volunteer (Brett et al., 2009). Collectively, selective toxicity of hesperidin in a male may be attributed to its sensitivity to longer gastric residence time and delayed absorption. Thus, the result of the present

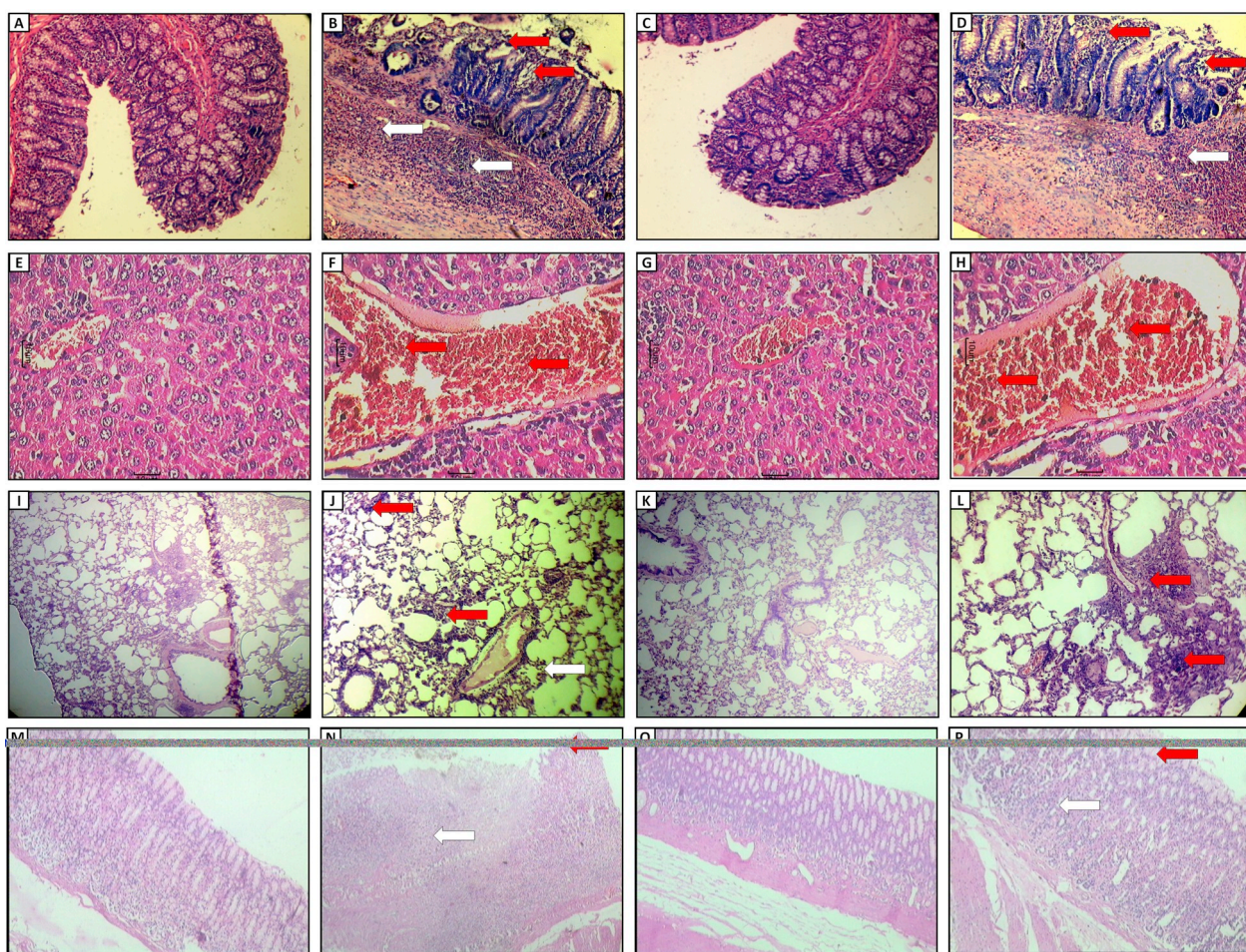


Fig. 2. Effect of hesperidin on the colon (A–D), liver (E–H), lung (I–L) and stomach (M–P) histology of male and female Sprague-Dawley rats during 13-weeks repeated dose toxicity study.

Photomicrographs of representative rats from respective groups: vehicle control, i.e., hesperidin (0 mg/kg) (A, E, I and M) for male, hesperidin (1000 mg/kg) treated (B, F, J, and N) for male, vehicle control, i.e., hesperidin (0 mg/kg) (C, G, K and O) for female and hesperidin (1000 mg/kg) (D, H, L and P) for female, (H&E stain) at 40X. Hesperidin was administered orally daily for 13-weeks.

investigation may provide some insights for the preparation of further suitable dosage forms of hesperidin for clinical application.

Since data obtained from acute toxicity is not sufficient to apply for clinical implication, sub-chronic toxicity studies have significant importance in determining the safety profile of drugs or chemical moieties (Aniagu et al., 2005). Although most of the plant-derived medicines showed low acute toxicity, repeated dose toxicity is required to ascertain the possible toxic effects of a substance likely to arise from repeated administration over a limited period. Hence, the present study was undertaken to investigate the effects of hesperidin after the oral administration in rats for 13-weeks. In the present investigation, chronic administration of hesperidin for 13-weeks did not show any significant changes in animal behavior, food and water consumptions as well as necropsy findings. However, a significant increase ($p < 0.05$ and $p < 0.01$) in body weight in hesperidin (1000 mg/kg) treated rats were found as compared to vehicle control rats. Body weight is considered as a simple and sensitive indicator of adverse effects after exposure to a substance (Raza et al., 2002; Teo et al., 2002). Increase in body weight of rats, after administration of hesperidin, could be due to alteration in appetite (not ascertained in the present study) and it also does not exert any deteriorative effect on the weight and growth of the animals. These findings are in agreement with the results of previous studies (Rider et al., 2014). Although an increase in the body weight of rats administered with hesperidin had been observed, this increased body weight was not associated with other clinical signs, and no related

indication of pathological abnormality was observed.

The analysis of hematology and clinical biochemistry parameters provide a slight peak into risk evaluation of hematological system alterations. The data obtained from the preclinical evaluation of drug toxicity have a higher predictive value for clinical toxicity (Olson et al., 2000). Repeated administration of hesperidin over 13-weeks caused minor but statistically significant hematological changes in red blood cell parameters which occurred in both male and female rats. It showed a significant increase in hemoglobin, red blood cell count, and hematocrit level. This increased hematological changes in red blood cell parameters may be due to the compensatorily increased hematopoiesis (Son et al., 2003; Udu et al., 2004). It has been reported that decreased MCHC value is associated with iron deficiency (Vijayalakshmi et al., 2000) and could be the cause of decreased hemoglobin concentration. Additionally, administration of hesperidin showed an increase in spleen weights. Histopathological evaluation of spleen also revealed evidence of inflammatory infiltration. However, these minor changes in red blood cell parameters, although statistically significant, were assumed to be toxicologically irrelevant because they were of minor magnitude or within current reference ranges for Sprague-Dawley rats of same sex and similar age (Alemáan et al., 1998; Han et al., 2010; Petterino and Argentino-Storino, 2006). These were neither dose-related nor reflected by any changes in other related parameters.

Serum biochemical parameters such as blood urea nitrogen, calcium, and sodium in hesperidin administered rats were significantly

altered after 13-weeks. These parameters are considered as the critical markers of kidney function (Tietz et al., 1994) and their biochemical determination is vital because kidney toxicity has been reported following the use of isolated products from herbal origin (Corns, 2003; Hilaly et al., 2004; Isnard Bagnis et al., 2004). Although these values were significantly altered, those were still within the normal range (Feres et al., 2006; Mitruka and Rawnsley, 1977; Palmeiro et al., 2003). Results of urinalysis showed no abnormal signs concerning the functioning of kidneys including urine volume, hematological and serum biochemistry tests, and necropsy. Besides the presence of mild inflammatory infiltration in the histopathological analysis of a kidney from hesperidin administered rat, they were devoid of any alteration which also supports these biochemical findings.

The consumption of medicinal plants without efficacy and safety studies can result in several side effects that may affect different organs. Administration of herbal drugs caused variation in blood glucose levels hence, there is a need to monitor it (Aniagu et al., 2005). In the present investigation, chronic administration of hesperidin showed a significant decrease in blood sugar level. However, this decrease in blood sugar level after administration of hesperidin was not dose-dependent and within the normal laboratory ranges. Similar findings were documented by a previous investigator where administration of flavonoid naringin induced alteration in blood sugar level which was not dose-dependent and within the normal laboratory ranges (Li et al., 2013). Additionally, histological assessment of pancreatic tissue from hesperidin treated rats also did not show any abnormalities which would reveal safety issues over the pancreatic tissue.

Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) are the hallmarks of liver function (Murray et al., 2009; Vijayalakshmi et al., 2000). ALT, which is a cytoplasmic enzyme present abundantly in the liver, get significantly increased during hepatocellular insult (Tennekoon et al., 1991). AST is an enzyme that is present extracellularly in various tissues like the heart, skeletal muscles, liver, kidneys, pancreas, and erythrocytes (Aniagu et al., 2004). AST is released from these cells in response to cellular damage or change in the membrane permeability of these cells. Hence, ALT is a more sensitive biomarker of hepatic damage than AST. Although values of AST and ALT in hesperidin treated group revealed significant difference as compared to the vehicle control group, these fluctuations were not sex or dose-dependent and were within the normal laboratory ranges (Li et al., 2013). Furthermore, slight histopathological abnormalities such as lymphocytic infiltration observed in the liver constitute of reversible changes. Animals administered with hesperidin also showed a significant increase in liver weight. Histological changes corroborated with these findings which were of low degree and reversible. Many researchers have reported lesions observed in the histopathological evaluation of liver tissue in hesperidin treated groups which could be due to spontaneous generation in 20-week-old Sprague-Dawley rats, and these nonspecific changes were considered to be unrelated to the test compound (Greaves, 2011; Mohr et al., 1992). Thus, in the present investigation, these changes were not considered to be of any toxicological significance with reference to exposure to hesperidin.

An elevated level of cholesterol, triglycerides, and bilirubin, as well as reduction in albumin, are other significant indicators of failure in hepatic function (Greaves, 2011; Han et al., 2011; Shah and Divyang Patel, 2010). Usually, fat accumulation occurs in hepatocytes during hepatic damage. However, in the present study, chronic administration of hesperidin did not show any significant change in the level of cholesterol. Moreover, in this case also, the histopathological examinations of the liver from hesperidin administered rats did not indicate any remarkable pathological changes including fatty changes in liver and obstruction in the intrahepatic bile duct. It has been documented that obstruction in intrahepatic bile duct resulted in significant rise in the level of cholesterol and if this problem persists, hypercholesterolemia may develop (Isnard Bagnis et al., 2004; Raza et al., 2002). Hence, no significant changes in cholesterol indicated that hesperidin is not

associated with fat metabolism. However, there was a significant increase in the level of triglyceride after administration of hesperidin, but it was not dose-dependent and was within the normal laboratory ranges (Li et al., 2013).

The results of acute and sub-chronic toxicity studies did not show any signs of toxicity attributable to the administration of hesperidin. Since, administration of oral dose of hesperidin (1000 mg/kg/day) for 13-weeks induces toxicological changes in biochemical, hematological, anatomical, and histopathological examination thus it can be defined as the Low Observed Adverse Effect Level (LOAEL) for rats of both sexes under the experimental conditions used. However, toxicity in humans cannot always be entirely extrapolated from animal studies and thus, clinical evaluation should be performed to precisely define the level of safe dosage in humans.

5. Conclusion

In summary, results of the present acute toxicity study showed that acute oral lethal dose of hesperidin was more than 5000 mg/kg with a median lethal dose (LD₅₀) value of 4837.5 mg/kg. The sub-chronic oral toxicity study suggested that oral administration of hesperidin had Low Observed Adverse Effect Level (LOAEL) at 1000 mg/kg for both male and female Sprague-Dawley Rats. Thus, hesperidin isolated from citrus fruits showed a good safety profile in animal study and can be considered as an important agent for clinical management of chronic disorders.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.yrtph.2019.04.001>.

Disclosure of interest

The authors declare that they have no conflicts of interest concerning this article.

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